



(USA) Principal, Data Engineer

[Apply](#) Sunnyvale, CA Full time Posted Yesterday R-2404738

[Position Summary...](#)

[What you'll do...](#)

Principal Data Engineer – Agentic Data Platforms

The Mission

As a **Principal Data Engineer**, you are the enterprise-scale technical authority responsible for shaping and evolving Sam's Club's data platform architecture to power intelligent, autonomous systems at scale.

Your mission is to architect and steward the foundational data systems that enable AI agents, copilots, and large-scale analytics to operate with reliability, semantic clarity, and enterprise-grade trust.

You do not simply build systems—you define how the organization builds systems.

About the Team

Sam's Club is a membership warehouse club serving millions of members with high-quality products and services at unmatched value. From Scan & Go and Club Pickup to Pharmacy, Travel, and Business Services, Sam's operates at enterprise scale across digital and physical channels.

The Data organization powers this ecosystem through scalable platforms that enable real-time decision-making, personalization, automation, and agentic AI innovation.

What You'll Do

Define Enterprise Data Architecture

- Architect and evolve the core data platform strategy across batch, streaming, and hybrid systems to support both traditional analytics and AI-native workloads.
- Establish and steward architectural standards that unify structured, semi-structured, and unstructured data across domains.

Lead Agentic Data Enablement

- Define enterprise patterns for agent-ready data, ensuring systems are discoverable, semantically rich, and optimized for LLMs, copilots, and multi-agent workflows.
- Shape reference architectures for RAG, real-time feature pipelines, vector indexing, and graph-augmented reasoning.

Drive Cross-Domain Platform Strategy

- Partner with Engineering, AI/ML, Product, and Platform leaders to align roadmaps with long-term autonomous decision-making goals.
- Influence and guide multiple teams in adopting scalable patterns without direct authority.

Architect for Trust, Reliability, and Scale

- Define standards for observability, telemetry, lineage, governance, and AI auditability across enterprise data systems.
- Design resilient, fault-tolerant systems that support millions of users and mission-critical retail operations.

Modernize Legacy Systems

- Lead modernization initiatives that transform traditional data lake systems into composable, event-driven, and agent-aware platforms.
- Balance short-term delivery with long-term scalability and maintainability.

Elevate Technical Excellence

- Mentor and guide Staff Engineers and Senior Engineers.
- Lead design reviews and steward architectural decisions across high-risk, high-impact initiatives.
- Raise the technical bar for the organization through principled engineering standards.

What You'll Bring

Enterprise-Scale Experience

- 10–15+ years building and evolving large-scale distributed data platforms.
- Proven track record of shaping architecture across multiple domains or organizations.

Deep Data Platform Expertise

- Strong experience in cloud-native ecosystems (GCP or Azure preferred), including BigQuery, Dataflow, Pub/Sub, or equivalent.
- Expertise across batch and streaming systems (Kafka, Spark Structured Streaming, Flink, Druid, etc.).
- Experience designing hybrid real-time + analytical architectures.

Agentic & AI-Ready Systems Knowledge

- Strong understanding of semantic modeling, embeddings, knowledge graphs, and vector search.
- Experience designing data systems that support RAG, context enrichment, and agent orchestration.
- Ability to reason about schema, latency, storage format, and their impact on AI reasoning quality.

Governance & Observability Leadership

- Advanced knowledge of data quality frameworks, access control, lineage, compliance, and auditability.
- Experience defining trust and safety standards for AI-enabled systems.

Strong Engineering Foundation

- Fluency in Python, Java, or Scala.
- Deep experience with Spark/PySpark and SQL optimization at scale.
- Strong systems thinking and performance optimization skills.

Influence Without Authority

- Demonstrated ability to align diverse stakeholders and drive architectural adoption across teams.
- Clear, executive-level communication and the ability to translate technical strategy into business value.

At Sam's Club, we offer competitive pay as well as performance-based bonus awards and other great benefits for a happier mind, body, and wallet!

-Health benefits include medical, vision and dental coverage

-Financial benefits include 401(k), stock purchase and company-paid life insurance

-**Paid time off benefits** include PTO, parental leave, family care leave, bereavement, jury duty, and voting. You will also receive PTO and/or PPTO that can be used for vacation, sick leave, holidays, or other purposes. The amount you receive depends on your job classification and length of employment. It will meet or exceed the requirements of paid sick leave laws, where applicable.

For information about PTO, see <https://one.walmart.com/notices>.

- **Other benefits** include short-term and long-term disability, company discounts, Military Leave Pay, adoption and surrogacy expense reimbursement, and more.

Live Better U is a company paid education benefit program for full-time and part-time associates in Walmart and Sam's Club facilities. Programs range from high school completion to bachelor's degrees, including English Language Learning and short-form certificates. Tuition, books, and fees are completely paid for by Walmart.

Eligibility requirements apply to some benefits and may depend on your job classification and length of employment. Benefits are subject to change and may be subject to a specific plan or program terms.

For information about benefits and eligibility, see [One.Walmart](#).

The annual salary range for this position is \$143,000.00 - \$286,000.00

Additional compensation includes annual or quarterly performance bonuses.

Additional compensation for certain positions may also include Regional Pay Zone (RPZ) (based on location).

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Minimum Qualifications...

Outlined below are the required minimum qualifications for this position. If none are listed, there are no minimum qualifications.

Option 1: Bachelor's degree in Computer Science and 5 years' experience in software engineering or related field.

Option 2: 7 years' experience in software engineering or related field. Option 3: Master's degree in Computer Science and 3 years' experience in software engineering or related field.

4 years' experience in data engineering, database engineering, business intelligence, or business analytics.

Preferred Qualifications...

Outlined below are the optional preferred qualifications for this position. If none are listed, there are no preferred qualifications.

Data engineering, database engineering, business intelligence, or business analytics, ETL tools and working with large data sets in the cloud, Master's degree in Computer Science or related field and 5 years' experience in software engineering or related field, We value candidates with a background in creating inclusive digital experiences, demonstrating knowledge in implementing Web Content Accessibility Guidelines (WCAG) 2.2 AA standards, assistive technologies, and integrating digital accessibility seamlessly. The ideal candidate would have knowledge of accessibility best practices and join us as we continue to create accessible products and services following Walmart's accessibility standards and guidelines for supporting an inclusive culture.

Primary Location...

809 11th Ave, Sunnyvale, CA 94089-4731, United States of America

Walmart and its subsidiaries are committed to maintaining a drug-free workplace and has a no tolerance policy regarding the use of illegal drugs and alcohol on the job. This policy applies to all employees and aims to create a safe and productive work environment.

About Us



About Walmart:

Fifty years ago, Sam Walton started a single mom-and-pop shop and transformed it into the world's biggest retailer. Since those founding days, one thing has remained consistent: our commitment to helping our customers save money so they can live better. Today, we're reinventing the shopping experience and our associates are at

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